

MAIN ARTICLE

System Dynamics Review and publications 1985–2017: analysis, synthesis and contributions

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Abstract

System Dynamics Review (SDR) is an important journal for the system dynamics field and its researchers. This article offers a systematic review of the system dynamics articles published in SDR and other related academic journals between 1985 and 2017. The study classifies the material by using a bibliometric analysis of 498 articles in SDR and 943 articles in other journals that are included in the Web of Science database. The results suggest that the most cited authors within SDR have published research in three broad areas: (i) model validity (confidence), tests, and other similar formal procedures; (ii) the modeling of dynamic problems of wide interest; and (iii) group model building. In other journals, an additional stream of system dynamics research proves to be highly influential. This stream focuses on better understanding and improving the process of dynamic decision making, addressing topics such as organizational learning and feedback misperceptions.

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Introduction

System Dynamics Review (SDR) is a valuable journal that has helped expand the system dynamics discipline worldwide. SDR was created after the 1984 System Dynamics (SD) Conference in Oslo, when David Andersen, in his role as the president of the newly formed System Dynamics Society, asked George Richardson to serve as the society's first Vice President for Publications. SDR exists to communicate to a wide audience the advances in the application of system dynamics perspectives and methods of societal, technical, managerial and environmental problems.

Although the International System Dynamics Society celebrated its 60th anniversary in 2017, few studies have systematically reviewed the publications of SD scholars in the field. For example, Forrester (2007a, 2007b) offers a personal view of the first 50 years of SD and then reflects upon the next 50 years of system dynamics. Similarly, Sterman (2018) offers his assessment of the first 60 years of the SD field and makes suggestions for the “path forward.” Richardson (1999) also presents his considerations regarding the foundations of SD. Other good examples of publications analysis within SDR

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were performed by Martinez-Moyano and Richardson (2013), Scholl (1995) and Richardson (1991).

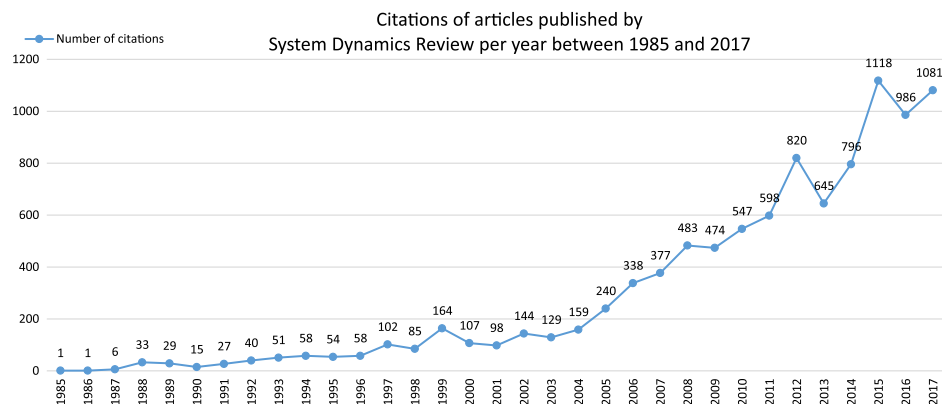
The main objective of this work is to explore the contribution that SDR has made to SD research and the most influential thematic works published in SDR since its establishment in 1985 until 2017. The second major objective is to perform an analysis of those SD publications that appeared in journals outside SDR between 1985 and 2017. This work should be of interest and importance to SD researchers. First, for readers, a review of SDR should be helpful because it will provide an overview of publications, citations, authors and topics of interest. Second, for people who have published articles on SD (or are thinking of doing so), this review serves as a guide, offering information on the most cited content, topics of interest, sources and authors. This information is valuable, especially for new SD researchers who are still trying to understand the nuances of the field. Finally, this study not only presents past publication trends but also attempts to reveal where SD offers potential areas of interest for follow-up research.

Overview of SD articles published between 1985 and 2017

This article collects bibliographic information from the Web of Science (WOS) database and selects only research (main) articles, short articles, and notes and insights from SDR and other journals published between 1985 and 2017. A total of 498 articles in SDR that matched such criteria were found. These articles received 9864 citations, for a rate of approximately 20.94 citations per study conducted. The number of citations remained steady between 1988 and 2001, with an average of 22 citations per study. After 2001, the number of citations experienced a significant upward trend, reaching 1118 citations in 2015. Figure 1 shows the dynamics of citations to papers between 1985 and 2017.

The most cited article between 1985 and 2017 was written by Barlas (1996), who discussed the formal aspects of model validity in SD, such as structural tests, structure-oriented behavior tests and behavior pattern tests. Barlas (1996) has 391 citations (Cit) and an average number of citations per year (ACit) of 17.8. The formal aspects of model testing and validity are one of the key topics covered by SDR—32 percent of the top 100 most cited articles in SDR between 1985 and 2017 cover the topics of formal validity, confidence and testing in SD. This research stream is focused on reflecting upon and responding to criticisms that SD does not employ formal, objective, quantitative procedures for testing and evaluation models, as is supposed to be fundamental to scientific inquiry (Barlas and Carpenter, 1990). The influential success of Barlas (1996) lies in its provision of formal validity procedures and tests to help SD experts explain their tacit knowledge. Other formal procedures published in SDR include the process of SD modeling

Fig. 1. Number of citations to papers published by *System Dynamics Review* between 1985 and 2017. Citations per year were collected from the Web of Science database [Color figure can be viewed at wileyonlinelibrary.com]



(Forrester, 1994), a method for collecting and analyzing qualitative data (Luna-Reyes and Andersen, 2003), the role of qualitative models in SD research (Coyle, 2000), model structure analysis (Oliva, 2004) and guidelines for simulation-based research in social sciences (Rahmandad and Sterman, 2012).

The second most cited SDR paper was written by Sterman (2002) (Cit: 257; ACit: 16.1), and it discusses how to learn about complex systems via SD simulation. This paper is also an example of a well-developed formal guide that facilitates the articulation of complex problems using feedback structures, simulating scenarios, improving scientific reasoning skills, strengthening the group process, and overcoming the defensive routines of individuals and teams. SD scholars have also developed several applications that enable people to understand and improve the actions embedded in complex systems. In fact, more than 57 percent of the 100 most cited articles in SDR between 1985 and 2017 are related to the modeling of dynamic problems of wide interest, such as global warming (Naill *et al.*, 1992; Sterman and Sweeney, 2002), sustainable development (Harich, 2010; Moxnes, 2000), drug use dynamics (Gardiner and Shreckengost, 1987; Homer, 1993; Coyle and Alexander, 1997), conservation policies (Ford and Bull, 1989), water scarcity (Fernandez and Selma, 2004) and disasters (Cooke, 2003; Owens *et al.*, 2011). The third most cited paper in SDR was written by Forrester (1994) (Cit: 180; ACit: 7.5), and it relates to both of the described streams in SDR: (i) formal procedures and (ii) the modeling and analysis of complex dynamic problems. Forrester (1994) also attempts to link SD with soft approaches (e.g. soft operations research and systems thinking) in the problem conceptualization stage of SD modeling.

Finally, the fourth most cited paper in SDR was written by Vennix (1999) (Cit: 177; ACit: 9.32), who discusses group model building. Group model building is the third most popular stream within SDR. This stream captures almost 10 percent of the top 100 most cited articles in SDR between 1985 and 2017. Group model building refers to an SD model building process in

which a client group is deeply involved in the process of model construction (Vennix, 1999). Richardson and Andersen (1995) also analyze how group model-building projects can support large client groups in business model formulation. Two years later, Andersen and Richardson (1997) extended previously reported works and presented scripts and techniques for developing group model-building sessions. Akkermans and Vennix (1997) highlight three factors experienced by clients to better explain the effects of group model building: communications (e.g. if people communicate openly and effectively in the course of the modeling process, they will develop a feeling of ownership of the resulting model), thoroughness that leads to usability (e.g. the better a model is, the greater the utility of its findings will be) and political sensitivity, which decreases the willingness to cooperate (e.g. politically sensitive problems affect the willingness of the team members to cooperate in the modeling process). For a complete review of the group model-building effectiveness topic, see Rouwette *et al.* (2002, 2011). Table 1 offers a complete list of the top 20 most cited articles in SDR.

Secondly, articles involving SD published outside SDR were drawn from the Thomson Reuters WOS database based on document relevance and the solid reputation of the source website. The WOS database was filtered for the following areas: engineering, management/business/economics, social science, computer science, mathematics, health care science, energy, ecology, planning development, urban studies, environmental studies and agronomy. The search used the following keywords: SD modeling, stock flow and causal loop. This selection criterion returned 1099 articles. A manual scan eliminated those articles that did not belong to the SD field. The elimination process produced a subset of 943 articles.

These 943 selected articles about SD modeling between 1985 and 2017 (excluding those published in SDR) cover a broad body of literature in the following areas: management/business (23 percent), environmental sciences (17.7 percent), civil/industrial engineering (11 percent), environmental engineering (8.9 percent), mathematical methods in social sciences (8.9 percent), operations research and management science (8.7 percent), water resources (6.6 percent), computer science (6.2 percent) and other categories (9 percent).

As can be seen in Table 2, the most cited SD article in the WOS database is entitled “Modeling managerial behavior—misperceptions of feedback in a dynamic decision-making experiment” (Sterman, 1989), with 1109 citations. The second most cited article published outside SDR is “System dynamics modeling for public health: Background and opportunities” (Homer and Hirsch, 2006), with 293 citations.

The third most cited article outside SDR is “Learning from evidence in a complex world” (Sterman, 2006), with 289 citations. The most cited articles beyond SDR are mostly related to understanding complex dynamic problems in areas such as managerial decisions, waste management, climate change and health management.

Table 1. The most cited articles in *System Dynamics Review* from 1985 to 2017

Ranking	Title	Author	Citations	Year	Citation/ year
1	Formal aspects of model validity and validation in system dynamics	Barlas Y	386	1996	19.30
2	All models are wrong: reflections on becoming a systems scientist	Sterman J	256	2002	18.29
3	System dynamics, systems thinking and soft OR	Forrester J	180	1994	8.18
4	Group model building: tackling messy problems	Vennix J	176	1999	10.35
5	Bathtub dynamics: initial results of a systems thinking inventory	Sweeney L, Sterman J	171	2000	10.69
6	Scripts for group model building	Andersen D, Richardson G	155	1997	8.16
7	Teamwork in group model building	Richardson G, Andersen D	147	1995	7.00
8	Dynamic modeling of product development processes	Ford D, Sterman J	135	1998	7.50
9	Systems thinking—critical thinking skills for the 1990s and beyond	Richmond B	135	1993	5.87
10	System dynamics applied to project management: a survey, assessment and directions for future research	Lyneis J, Ford D	114	2007	12.67
11	Group model building: adding more science to the craft	Andersen D, Richardson G, Vennix J	106	1997	5.58
12	Using system dynamics to improve public participation in environmental decisions	Stave K	100	2002	7.14
13	Expert knowledge elicitation to improve formal and mental models	Ford D, Sterman J	98	1998	5.44
14	Cloudy skies: assessing public understanding of global warming	Sterman J, Sweeney L	95	2002	6.79
15	Strategic management of complex projects: a case study using system dynamics	Lyneis J, Cooper K, Els S	87	2001	5.80
16	Philosophical roots of model validation: two paradigms	Barlas Y, Carpenter S	87	1990	3.35
17	Problems with causal loop diagrams	Richardson G	82	1986	2.73
18	Collecting and analyzing qualitative data for system dynamics: methods and models	Luna-Reyes L, Andersen D	81	2003	6.23
19	Qualitative and quantitative modelling in system dynamics: some research questions	Coyle G	77	2000	4.81
20	Towards the definition and use of a core set of archetypal structures in system dynamics	Wolstenholme E	74	2003	5.69

Bibliometric method

This study classifies the material using a bibliometric approach (Merigo *et al.*, 2015). Bibliometrics is a research discipline that studies bibliographic material quantitatively (Broadus, 1987) and provides a general overview of a research topic according to a wide range of measures. This study uses VOS viewer software to draw a network map of the most cited articles based on bibliographic coupling (van Eck and Waltman, 2010). Bibliographic coupling

Table 2. The most cited articles in system dynamics modeling excluding SDR from 1985 to 2017

Rank	Title	Author	Source	Citations	Year	Citation/ year
1	Modeling managerial behavior—misperceptions of feedback in a dynamic decision-making experiment	Sterman J	<i>Management Science</i>	1109	1989	36.97
2	System dynamics modeling for public health: background and opportunities	Homer J, Hirsch G	<i>American Journal of Public Health</i>	293	2006	22.54
3	Learning from evidence in a complex world	Sterman J	<i>American Journal of Public Health</i>	289	2006	22.23
4	System dynamics modeling: tools for learning in a complex world	Sterman J	<i>California Management Review</i>	256	2001	14.22
5	Misperceptions of feedback in dynamic decision-making	Sterman J	<i>Organizational Behavior and Human Decision Processes</i>	249	1989	8.3
6	Effects of feedback complexity on dynamic decision-making	Diehl E, Sterman J,	<i>Organizational Behavior and Human Decision Processes</i>	186	1995	7.75
7	Capability traps and self-confirming attribution errors in the dynamics of process improvement	Repenning N, Sterman J	<i>Administrative Science Quarterly</i>	177	2002	10.41
8	Forecasting municipal solid waste generation in a fast-growing urban region with system dynamics modeling	Dyson B, Chang N	<i>Waste Management</i>	167	2005	11.93
9	Cutting corners and working overtime: quality erosion in the service industry	Oliva R, Sterman J	<i>Management Science</i>	165	2001	9.17
10	Boom, bust, and failures to learn in experimental markets	Paich M, Sterman J	<i>Management Science</i>	159	1993	6.12
11	Transition challenges for alternative fuel vehicle and transportation systems	Struben J, Sterman J	<i>Environment and Planning B—Planning and Design</i>	155	2008	14.09
12	A system dynamics model to facilitate public understanding of water management options in Las Vegas, Nevada	Stave KA	<i>Journal of Environmental Management</i>	152	2003	9.5
13	Unanticipated side effects of successful quality programs: Exploring a paradox of organizational improvement	Sterman J, Repenning N, Kofman F	<i>Management Science</i>	144	1997	6.55
14	Economics—risk communication on climate: mental models and mass balance	Sterman J	<i>Science</i>	141	2008	12.82
15	Navigating towards sustainable development: a system dynamics approach	Hjorth P, Bagheri, A	<i>Futures</i>	134	2006	10.31
16	Understanding public complacency about climate change: adults' mental models of climate change violate conservation of matter	Sterman J, Booth Sweeney, L	<i>Climatic Change</i>	132	2007	11

Table 2. Continued

Rank	Title	Author	Source	Citations	Year	Citation/ year
17	Multiple tests for validation of system dynamics type of simulation-models	Barlas Y	<i>European Journal of Operational Research</i>	130	1989	4.64
18	System dynamics modeling of reservoir operations for flood management	Ahmad S, Simonovic S	<i>Journal of Computing in Civil Engineering</i>	122	2000	6.42
19	Why do not well-educated adults understand accumulation? A challenge to researchers, educators, and citizens	Cronin M, Gonzalez C, Sterman J	<i>Organizational Behavior and Human Decision Processes</i>	117	2009	11.7
20	System dynamics modeling for urban energy consumption and CO ₂ emissions: a case study of Beijing, China	Feng Y, Chen S, Zhang L	<i>Ecological Modelling</i>	114	2013	19

occurs when two documents cite a third common study (Broadus, 1987), thus establishing a similarity relationship between the two articles (and also implying interrelated research). The VOS viewer can perform clustering analysis based on bibliographic coupling, co-citation and co-authorship. I use bibliographic coupling analysis, which calculates the number of clusters by minimizing the distance between two articles.¹ In this clustering analysis, the distance between two articles indicates the approximate relatedness of the articles in terms of bibliographic links. In general, the closer two articles are located to each other, the stronger their relatedness. When two articles are connected via bibliographic coupling, the software establishes a link and calculates a weight based on the number of bibliographic couplings between these two articles; the software then maps the clusters using different colors (van Eck and Waltman, 2010). When several papers are located within the same cluster, it implies that they are discussing a very similar topic or content. The number of citations, by contrast, determines the size of the circle of an item (e.g. article or journal). The higher the number of citations of an item is, the larger the circle of that item.

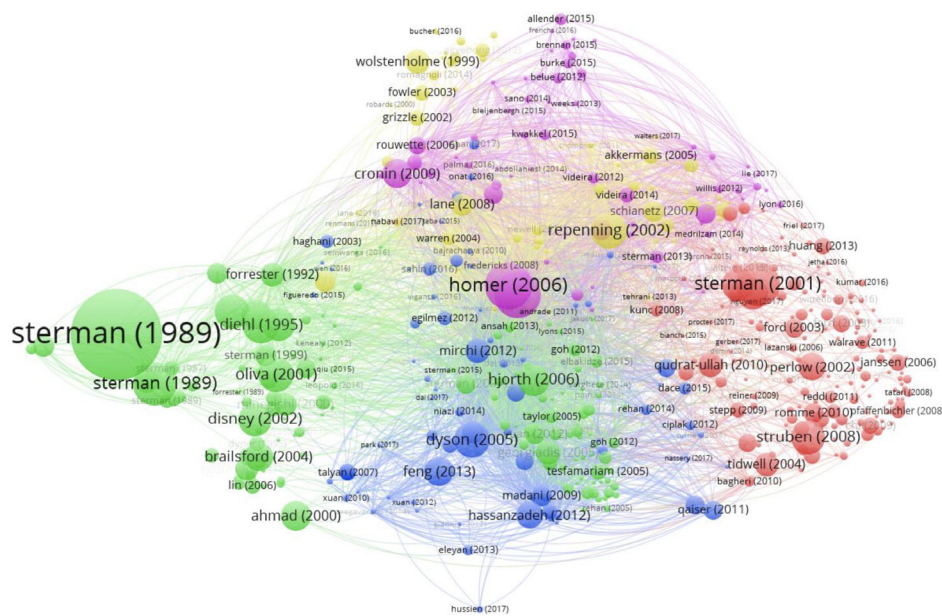
Bibliometric results

Networks among articles published in SDR between 1985 and 2017

Figure 2 summarizes the bibliographic coupling analysis of SDR by article. This analysis shows the number of citations of an article through the size of a node and the links that appear when bibliographic coupling occurs.

ⁱvan Eck and Waltman (2010) describe the objective function to be minimized and its restrictions that allow the VOS viewer software to map out all clusters (see p. 532).

Fig. 3. Bibliographic coupling network map based on articles published outside *System Dynamics Review*. The map shows 500 articles with the highest total link strength between 1985 and 2017 [Color figure can be viewed at wileyonlinelibrary.com]



diagramming methods employed in SD, with a focus on how to communicate the content of SD models. The article with the second highest number of bibliographic coupling links outside SDR is “Why has feedback systems thinking struggled to influence strategy and policy formulation? Suggestive evidence, explanations and solutions” (Warren, 2004), with 1096 links. The article with the third highest coupling is “Forecasting municipal solid waste generation in a fast-growing urban region with system dynamics modeling” (Dyson and Chang, 2005), with 878 links. Figure 3 details five clusters of articles based on a single bibliographic coupling. Although all clusters have a highly cited hub paper connecting articles to broad-interest general applications, the green cluster represents articles that focus mainly on understanding and improving decision-making processes, such as articles on the misperception of feedback loops (Stermann, 1989), organizational learning (Diehl and Stermann, 1995), path dependence (Stermann and Wittenberg, 1999), quality erosion (Oliva and Stermann, 2001), nonlinearity (Forrester, 1987), boom and bust dynamics (Paich and Stermann, 1993) and sustainable development (Hjorth and Bagheri, 2006). Significant topics of interest in other clusters deal with health and disease management (Homer and Hirsch, 2006; Stermann, 2006; Brennan *et al.*, 2015) (purple cluster), waste, pollution and water resource management (Dyson and Chang, 2005; Madani and Marino, 2009; Hassanzadeh *et al.*, 2012; Feng *et al.*, 2013) (blue cluster); climate change, energy and transport (Stermann and Sweeney, 2007; Struben and Stermann, 2008; Qudrat-Ullah and Seong, 2010) (red cluster), and ecology

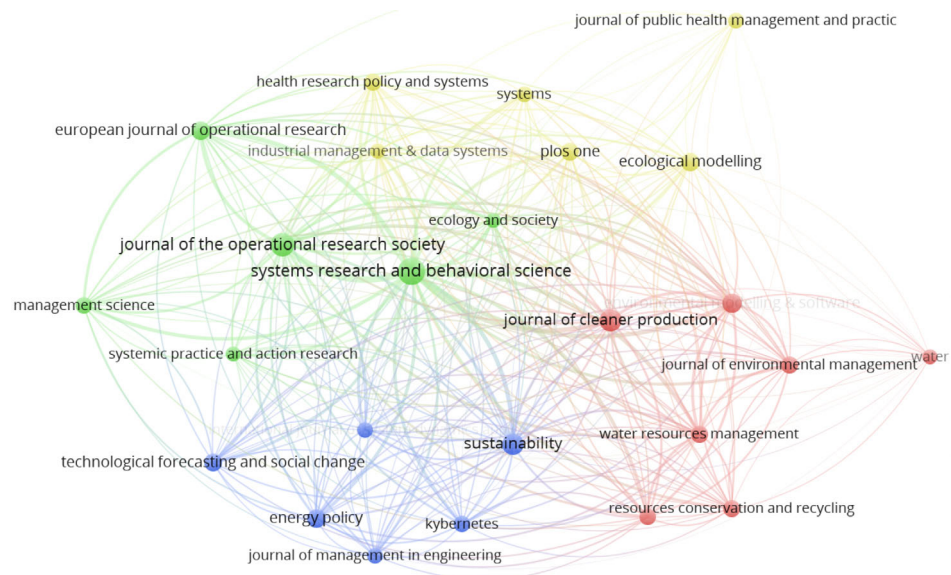
(Nguyen *et al.*, 2011; Sedlacko *et al.*, 2014) (yellow cluster). The yellow cluster is the only group that involves three hub articles; these analyze the process of improvement (Repenning and Sterman, 2002), qualitative and quantitative modeling (Wolstenholme, 1999) and SD diagramming usage (Lane, 2008).

Finally, Figure 4 shows the most influential journals (excluding SDR), being those that published more than five SD articles between 1985 and 2017. This network map displays four clusters of journals based on bibliographic coupling. The cluster analysis suggests four streams of SD research: (i) operational research and management science applications (green cluster), (ii) environmental analysis (red cluster), (iii) health management applications (yellow cluster) and (iv) general applications of SD modeling (blue cluster). Between 1985 and 2017, *System Research and Behavioral Science Journal* published the largest number of SD articles (35 articles), followed by *Journal of the Operational Research Society* (22 articles) and *Journal of Cleaner Production* (18 articles). However, the SD articles published in *Management Science* have the highest number of citations (1761).

Concluding remarks and future research

According to Ramos-Rodriguez and Ruiz-Navarro (2004), the intellectual basis upon which a discipline is developed is largely revealed by the

Fig. 4. Bibliographic coupling network map based on journals (excluding *System Dynamics Review*) that published more than five articles between 1985 and 2017 [Color figure can be viewed at wileyonlinelibrary.com]



citations of researchers' writings. Although some studies have described the state and evolution of SD research (Alfeld, 1995; Richardson, 1999; Forrester, 2007a, 2007b), no bibliometric studies have attempted to quantify and address the structure of SD research published in SDR and beyond this journal.

This article highlights the significant role of SDR in the development of the SD field and shows that the most cited articles have contributed to three major areas: modeling dynamic issues of wide appeal; model validity, testing and formal procedures; and group model building. Modeling dynamic problems of wide interest is the major stream of research and a common thread in SDR and beyond-SDR journals, involving applications in ecosystems, health care, supply chains, waste management, transport, climate change and energy. Another major stream of research in beyond-SDR journals focuses on better understanding and improving dynamic decision-making processes. This second stream has been successful in communicating the rationale behind SD models to a non-SDR audience (Morecroft, 1985) and formulating the causal and feedback structures behind "real" world dynamic problems (Richardson, 1986). This stream has also facilitated the modeling of explicit policy statements or rules that govern decision making (Forrester, 1992).

Although the analysis conducted here is based on a single bibliometric assumption (that references are a reliable indicator of the weights of certain data sources within the works of these authors), the bibliometric results reveal strong connections between foundational SD articles and applications. Cluster analysis seems to suggest three areas for future research and potential publication.

1. *Behavioral studies.* The analysis of why and how people misperceive and misunderstand the dynamics of a feedback system has been the most cited topic in SDR as well as in other journals. Sterman (1989) suggests that people do not properly perceive the structures behind the dynamics of systems (misperception) when they observe and manage such systems. Although articles in SDR have offered suggestions regarding sources that can reduce the misperceptions in various systems, such as markets (Kampmann and Sterman, 2014), competition between resource users (Moxnes, 2000), information processing (Kleinmuntz, 1993) or human-computer interfaces (Howie *et al.*, 2000), future research should more thoroughly explore the underlying mechanisms or routines that cause misperceptions among decision makers. Research in this area will not only support SD research but also contribute to the current debate about the dynamic capabilities within the strategy arena (Teece, 2007; Eisenhardt *et al.*, 2010; Helfat and Peteraf, 2015). SD research also has the potential to illuminate the current debate about the microfoundations of dynamic capabilities at the individual (Helfat and Peteraf, 2015) as well as organizational level (Barreto, 2010). SD researchers could and probably should look into and adopt tools and approaches from other related sciences, such as applying contributions from cognitive neuroscience to analyze managers' feedback misperceptions.
2. *Measuring the effects of interventions in the long term.* Research in SDR has shown that SD models developed after group-building interventions can support the analysis of organizational learning processes (Scholz *et al.*, 2015). Hence SD research can and should measure the effectiveness of these protocols in terms of the behaviors of decision makers not only

immediately after an intervention but also after some time has passed. Few studies in SDR have demonstrated the efficacy of SD interventions using simulations (Morecroft *et al.*, 1991; Cavaleri and Sterman, 1997; Scott *et al.*, 2013). Future research can determine key success factors to facilitate interventions, such as the length of interventions; the layout configuration; the background, experience and age of participants; the language of the facilitator; the emotional state of participants; and the protocols followed before, during and after interventions.

3. *Enhancing elicitation procedures for non-SD participants.* In most SD projects, the experts sketch out the causal, delay and feedback effects in consultation with management teams, either on a whiteboard or directly with computer software (Ackermann *et al.*, 2010; Torres *et al.*, 2017). An area of future research could explore how decision makers (e.g. managers and top management teams) can formulate stock and flow diagrams that are consistent with the principles of SD modeling (Forrester, 1994) while at the same time creating scenarios based on their business assumptions. Interactive modeling has been shown to be useful as a means of avoiding the development of linear and static perspectives in SD (Kampmann and Sterman, 2014), but there is less evidence indicating how to provide decision makers (lacking an engineering/systemic background) with a procedure to develop their own reliable simulation models (Andersen *et al.*, 2012).

This study has some limitations related to its research design and bibliometric approach. The main drawbacks of the research design are the analysis of data from a relatively short period between 1985 and 2017 and from a single database (WOS). This sample limits the potential scope of the obtained results because they represent only a portion of all SD publications. Further, a more detailed and comprehensive bibliometric analysis could be carried out on the publication data, yielding perhaps alternative networks and bibliometric couplings in different dimensions. This study is an initial step that aims to spark further research in this direction.

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Biography

Juan Pablo is an Assistant Professor of Strategy at the School of Economics and Business of the University of Chile, where he teaches strategic management. Dr. Torres holds a PhD in management from the Warwick Business School of the University of Warwick, an MSc in industrial engineering, and an undergraduate degree in industrial civil engineering from the University of Santiago of Chile. His research interests are in the area of strategic management, managerial decision making and systems modeling. His research

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